$Fibonacci\ Series\ -\ Precise\ Time\ Complexity$

$$T(n) = \begin{cases} 1 & if(n = 0) \\ 1 & if(n = 1) \\ T(n-1) + T(n-2) & if(n > 1) \end{cases}$$

$$T(n) = T(n-1) + T(n-2) + O(1)\ or, T(n-1) + T(n-2) + 1$$

$$\Rightarrow O(2^n)\ [This\ is\ also\ not\ precise\ or\ exact\ .]$$

$$T(n) = T(n-1) + T(n-2) + 1$$

$We\ split\ into\ two\ parts$:

$Homogeneous\ part$:

$$T(n) = T(n-1) + T(n-2)$$

$Non - Homogeneous\ part$: $+1$

$T(n) = T_h(n) + T_p(n), where\ , T_h(n) = T(n-1) + T(n+2)\ and$

$T_p(n) = 1\ or\ C[i.e. Constant]$

$\boldsymbol{Now, solve, Homogeneous\ Part:}\ \boldsymbol{T(n) = T(n-1) + T(n-2)}$

$\boldsymbol{Assuming\ T(n) = x^n}$

$\boldsymbol{Why\ this\ guess}$?
$\boldsymbol{Because}$:

- $\boldsymbol{Recurrences\ often\ behave\ like\ exponentials}$
- $\boldsymbol{Similar\ to\ how\ linear\ differential\ equations\ use\ e^x}$

$\boldsymbol{Therefore,\quad we\ replace\ every\ term\ as}$:

$$\boldsymbol{T(n) = T(n-1) + T(n-2)}$$

$$\boldsymbol{x^n = x^{n-1} + x^{n-2}}$$

we want to *eliminate the dependence on* `$\boldsymbol{n}$` ***and get a*** *constant equation in* $\boldsymbol{x}$.

We want to convert everything into the ***same power base level***.

So we divide by the ***smallest power present***, *which is*: x^{n-2}

$\boldsymbol{Dividing\ both\ side\ with\ `x^{n-2}`\ we\ get}$:

$$\boldsymbol{\frac{x^n}{x^{n-2}} = \frac{x^{n-1}}{x^{n-2}} + \frac{x^{n-2}}{x^{n-2}}}$$

$$\boldsymbol{\Longrightarrow x^{n-(n-2)} = x^{n-1-(n-2)} + x^{n-2-(n-2)}}$$

$$\boldsymbol{\Longrightarrow x^2 = x^1 + x^0}$$

$$\boldsymbol{\Longrightarrow x^2 = x + 1}$$

$$\boldsymbol{\Longrightarrow x^2 - x - 1 = 0}$$

$\boldsymbol{By\ Quadratic\ Formula}$:

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$\boldsymbol{Where, a = 1\ , b = -1\ and\ c = -1}$

$$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \times 1 \times -1}}{2 \times 1}$$

$$= \frac{1 \pm \sqrt{1 + 4}}{2}$$

$$= \frac{1 \pm \sqrt{5}}{2}$$

$\boldsymbol{So\ we\ get}$:

$$\varphi = \frac{1 + \sqrt{5}}{2} = \frac{1 + 2.236}{2} = \frac{3.236}{2} \approx 1.618$$

$$\psi = \frac{1 - \sqrt{5}}{2} = \frac{1 - 2.236}{2} = \frac{-1.236}{2} \approx -0.618$$

$\boldsymbol{The\ general\ solution\ for\ the\ homogeneous\ part\ is\ a\ linear\ combination\ of\ these\ roots}$:

$$T_h(n) = A\varphi^n + B\psi^n\ [where, A\ and\ B\ are\ coefficients]$$

$As, \psi^n =$

$(-0.618)^2 = 0.382$

$(-0.618)^3 = -0.236$

$(-0.618)^4 = 0.145$

$(-0.618)^5 = -0.090$

……

$Hence\ we\ can\ tell:\ \psi^n \to 0$.

$Where\ , \varphi^n = (1.618)^n \to \infty$

$$Solving\ , Non - homogeneous\ part:\ +1$$

$Since\ the\ non - homogeneous\ part\ is\ a\ constant\ (which\ is\ essentially\ 1 \times 1^n),$
$and\ 1\ is\ not\ one\ of\ our\ roots.$

$As\ roots\ were:$

$$\varphi = \frac{1+\sqrt{5}}{2} = \frac{1+2.236}{2} = \frac{3.236}{2} \approx 1.618$$

$$\psi = \frac{1-\sqrt{5}}{2} = \frac{1-2.236}{2} = \frac{-1.236}{2} \approx -0.618$$

$, we\ can\ guess\ that\ the\ particular\ solution\ is\ just\ some\ constant\ C$:

$$T_p(h) = C$$

$Substitute\ `C`\ back\ into\ the\ original\ recurrence[T(n) = T(n-1) + T(n-2) + 1]$:

$$\Rightarrow C = C + C + 1$$
$$\Rightarrow C = 2C + 1$$

$$\Rightarrow C = -1$$

$Now,$

$$T(n) = T_h(n) + T_p(n)$$

$$T(n) = A\varphi^n + B\psi^n - 1$$

$$T(n) = A\left(\frac{1+\sqrt{5}}{2}\right)^n + B\left(\frac{1-\sqrt{5}}{2}\right)^n - 1$$

$$Or, T(n) = c_1\left(\frac{1+\sqrt{5}}{2}\right)^n + c_2\left(\frac{1-\sqrt{5}}{2}\right)^n - 1$$

$Here\ A\ or\ c_1\ and\ B\ or\ c_2\ are\ constants\ or\ coefficients.$

$Though\ , calculation\ of\ these\ A\ or\ c_1\ and\ B\ or\ c_2\ are\ not\ needed,$ $then\ also\ we\ can\ calculate\ them:$

$We\ know\ base\ cases,$

```
    if (n == 0 || n == 1) // Base case
    {
        return n; // If n is 0 or 1, return n
    }
```

$i.e. T(0) = 1\ and\ T(1) = 1\ .$

$$\boldsymbol{For\ n = 0}:$$

$$\boldsymbol{T(n) = c_1(\varphi)^n + c_2(\psi)^n - 1}$$
$$\boldsymbol{T(0) = c_1(\varphi)^0 + c_2(\psi)^0 - 1}$$
$$\boldsymbol{1 = c_1 \times 1 + c_2 \times 1 - 1}$$
$$\boldsymbol{\Longrightarrow c_1 \times 1 + c_2 \times 1 - 1 = 1}$$
$$\boldsymbol{\Longrightarrow c_1 + c_2 = 2}$$

$$\boldsymbol{For\ n = 1}:$$

$$\boldsymbol{T(n) = c_1(\varphi)^n + c_2(\psi)^n - 1}$$
$$\boldsymbol{T(1) = c_1(\varphi)^1 + c_2(\psi)^1 - 1}$$
$$\boldsymbol{\Longrightarrow 1 = c_1\varphi + c_2\psi - 1}$$
$$\boldsymbol{\Longrightarrow c_1\varphi + c_2\psi = 2}$$

$$\boldsymbol{Now\ , c_1 + c_2 = 2 \Longrightarrow c_2 = 2 - c_1}$$
$$\boldsymbol{\therefore c_1\varphi + (2 - c_1)\psi = 2}$$
$$\boldsymbol{\Longrightarrow c_1\varphi + (2 - c_1)\psi = 2}$$
$$\boldsymbol{\Longrightarrow c_1\varphi + 2\psi - c_1\psi = 2}$$
$$\boldsymbol{\Longrightarrow c_1\varphi - c_1\psi + 2\psi = 2}$$
$$\boldsymbol{\Longrightarrow c_1(\varphi - \psi) = 2 - 2\psi}$$

$$\boldsymbol{Now, recall\ the\ actual\ values\ of\ our\ roots\ from\ the\ characteristic\ equation}:$$

$$\boldsymbol{\varphi = \frac{1 + \sqrt{5}}{2}}$$

$$\boldsymbol{\psi = \frac{1 - \sqrt{5}}{2}}$$

$$\Rightarrow c_1\left(\frac{1+\sqrt{5}}{2}-\frac{1-\sqrt{5}}{2}\right)=2-2\left(\frac{1-\sqrt{5}}{2}\right)$$

$$\Rightarrow c_1\left(\frac{1+\sqrt{5}-1+\sqrt{5}}{2}\right)=2-\left(1-\sqrt{5}\right)$$

$$\Rightarrow c_1\left(\frac{2\sqrt{5}}{2}\right)=2-1+\sqrt{5}$$

$$\Rightarrow c_1\left(\sqrt{5}\right)=1+\sqrt{5}$$

$$\Rightarrow c_1=\frac{1+\sqrt{5}}{\sqrt{5}}$$

$Now\ that\ we\ have\ c_1, we\ can\ easily\ find\ c_2$:

$$\Rightarrow c_2=2-c_1$$

$$\Rightarrow c_2=2-\left(\frac{1+\sqrt{5}}{\sqrt{5}}\right)$$

$$\Rightarrow c_2=\frac{2\sqrt{5}-1-\sqrt{5}}{\sqrt{5}}$$

$$\Rightarrow c_2=\frac{-1+\sqrt{5}}{\sqrt{5}}$$

$$or, c_2=\frac{\sqrt{5}-1}{\sqrt{5}}$$

$Therefore\ , full\ equation\ stand\ as$:

$$T(n) = A\varphi^n + B\psi^n - 1$$

$$T(n) = A\left(\frac{1+\sqrt{5}}{2}\right)^n + B\left(\frac{1-\sqrt{5}}{2}\right)^n - 1$$

$$Or, T(n) = c_1\left(\frac{1+\sqrt{5}}{2}\right)^n + c_2\left(\frac{1-\sqrt{5}}{2}\right)^n - 1$$

$Or,$

$$T(n) = \left(\frac{1+\sqrt{5}}{\sqrt{5}}\right) \times \left(\frac{1+\sqrt{5}}{2}\right)^n + \left(\frac{\sqrt{5}-1}{\sqrt{5}}\right) \times \left(\frac{1-\sqrt{5}}{2}\right)^n - 1$$

$As, \psi^n \to 0$ *and* $\varphi^n \to \infty$ *and* -1 *is neglible due to constant* ,

Also , $c_1 = \frac{1+\sqrt{5}}{\sqrt{5}}$ *and* $c_2 = \frac{\sqrt{5}-1}{\sqrt{5}}$ *are both constants and neglible*:

Therefore , *our time complexity*: $\Theta(\varphi^n) \approx \Theta(1.618^n)$